

Sub.: _____

SAT	SUN	MON	TUE	WED	THU	FRI
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Binary Cross Entropy (BCE) Loss Function:

$$L = - [y \log a + (1-y) \log (1-a)]$$

where :-

 y = actual target a = predicted probability

$$\frac{dL}{da} = \frac{d}{da} [-y \log a + (1-y) \log (1-a)]$$

$$= \frac{d}{da} [-y \log a] + \frac{d}{da} [(1-y) \log (1-a)]$$

$$= -y \cdot \frac{1}{a} + [-(1-y) \cdot \frac{1}{1-a} \cdot \frac{d}{da} (1-a)]$$

$$= -\frac{y}{a} + [-(1-y) \cdot \frac{1}{1-a} \cdot -1]$$

$$= -\frac{y}{a} + \frac{1-y}{1-a}$$

$$= \frac{-y(1-a) + a(1-y)}{a(1-a)}$$

$$= \frac{-y + ay + a - ay}{a(1-a)}$$

$$= \frac{a-y}{a(1-a)}$$

The sigmoid output, $a = \frac{1}{1 + e^{-z}}$

$$a = \frac{1}{1 + e^{-z}} = (1 + e^{-z})^{-1}$$

$$\frac{da}{dz} = \frac{d}{dz} (1 + e^{-z})^{-1}$$

$$= -1 (1 + e^{-z})^{-2} \cdot \frac{d}{dz} (1 + e^{-z})$$

$$= \frac{e^{-z}}{(1 + e^{-z})^2} [(0-1) \cdot 1 + 1 \cdot 0] = \frac{-e^{-z}}{(1 + e^{-z})^2}$$

$$= \frac{e^{-z}}{(1 + e^{-z})^2} \cdot \frac{1}{(1 + e^{-z})} = \frac{e^{-z}}{(1 + e^{-z})^3}$$

$$= a \cdot \left[1 - \frac{1}{(1 + e^{-z})} \right] = a \cdot \left[1 - \frac{1}{1 + a} \right]$$

$$= a (1 - a)$$

$$\therefore \frac{da}{dz} = a (1 - a)$$

Applying chain rule :-

$$\frac{dL}{dz} = \frac{dL}{da} \times \frac{da}{dz}$$

$$= \frac{a - y}{a(1-a)} \times a(1-a)$$

$$= a - y$$

$$\therefore \frac{dL}{dz} = a - y$$